

Bias-Corrected Variance Estimator for Pro-Active Inference

(Intended as §2.5 + Appendix material)

Notation and Setup

We use the notation of Theorem ?? throughout. Recall that the PAI estimator is $\hat{\theta}_n = \frac{1}{n} \sum_{t=1}^n \phi_t$ with martingale increments $\Delta_t = \phi_t - \theta$, and the variance estimator is $\hat{\sigma}_n^2 = \hat{A}_n - \hat{B}_n$, where

$$\begin{aligned}\hat{A}_n &:= \frac{1}{nN^2} \sum_{t=1}^n \frac{\left(y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})\right)^2}{q_t(I_t)^2}, \\ \hat{B}_n &:= \frac{1}{nN^2} \sum_{t=2}^n \left(N\hat{\theta}_{t-1} - \sum_{i \in O_{t-1}} y_i - \sum_{i \notin O_{t-1}} \hat{f}^{(t-1)}(x_i) \right)^2.\end{aligned}$$

We also recall the population-level quantities that arise from the conditional variance decomposition (cf. proof of Theorem ??):

$$\begin{aligned}A_n &:= \frac{1}{nN^2} \sum_{t=1}^n \sum_{i \notin O_{t-1}} \frac{\left(y_i - \hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}, \\ B_n &:= \frac{1}{nN^2} \sum_{t=1}^n \left(\sum_{i \notin O_{t-1}} \left(y_i - \hat{f}^{(t-1)}(x_i)\right) \right)^2,\end{aligned}$$

so that $\frac{1}{n} \sum_{t=1}^n \text{Var}(\Delta_t \mid \mathcal{F}_{t-1}) = A_n - B_n \xrightarrow{P} \sigma^2$. Define the per-step plug-in estimate

$$\psi_t := \frac{1}{N} \left(\sum_{i \in O_{t-1}} y_i + \sum_{i \notin O_{t-1}} \hat{f}^{(t-1)}(x_i) \right), \quad (1)$$

so that $\phi_t = \psi_t + \zeta_t$ and $\Delta_t = \psi_t + \zeta_t - \theta$, where we denote the normalized AIPW correction

$$\zeta_t := \frac{1}{N} \frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}. \quad (2)$$

Note that ψ_t is $\mathcal{F}_{t-1}$ -measurable, while ζ_t introduces the only new randomness at step t . We have $\mathbb{E}[\zeta_t \mid \mathcal{F}_{t-1}] = \frac{1}{N} \sum_{i \notin O_{t-1}} (y_i - \hat{f}^{(t-1)}(x_i)) = \theta - \psi_t$.

2.5 Bias-Corrected Variance Estimator

Although $\hat{\sigma}_n^2 = \hat{A}_n - \hat{B}_n$ is a consistent estimator for σ^2 (Theorem ??), it exhibits a systematic negative finite-sample bias: $\hat{B}_n$ uses the running estimate $\hat{\theta}_{t-1}$ in place of the unknown true mean θ , causing $\hat{B}_n$ to overestimate B_n on average. This bias manifests as confidence intervals that are narrower than intended, degrading empirical coverage below the nominal $1 - \alpha$ level—especially at larger sampling fractions n/N where it accumulates over more rounds. In this section, we construct a correction term $\hat{C}_n$ that targets the dominant source of this bias, yielding a corrected estimator $\hat{\sigma}_{\text{cor}}^2 := \hat{A}_n - \hat{B}_n + \hat{C}_n$ that is slightly conservative.

Step 1: Exact decomposition of the bias

The following proposition, which is an algebraic consequence of the proof of Theorem ??, identifies the exact (pathwise) decomposition of $\hat{B}_n - B_n$.

Proposition 1 (Exact bias decomposition). *For every realization,*

$$\hat{B}_n - B_n = T_1 + T_2 - T_3, \quad (3)$$

where

$$T_1 := \frac{1}{n} \sum_{t=2}^n \left(\hat{\theta}_{t-1} - \theta \right)^2, \quad (4)$$

$$T_2 := \frac{2}{n} \sum_{t=2}^n (\theta - \psi_t) \left(\hat{\theta}_{t-1} - \theta \right), \quad (5)$$

$$T_3 := \frac{1}{n} (\theta - \psi_1)^2. \quad (6)$$

Proof. We recall from the proof of Theorem ?? that

$$B_n = \frac{1}{n} \sum_{t=1}^n (\theta - \psi_t)^2 = \frac{1}{n} \sum_{t=2}^n (\theta - \psi_t)^2 + \frac{1}{n} (\theta - \psi_1)^2. \quad (7)$$

Meanwhile, $\hat{B}_n = \frac{1}{n} \sum_{t=2}^n (\hat{\theta}_{t-1} - \psi_t)^2$ (note: the sum starts at $t = 2$). For each $t \geq 2$, the identity $a^2 - b^2 = (a - b)^2 + 2b(a - b)$ with $a = \hat{\theta}_{t-1} - \psi_t$ and $b = \theta - \psi_t$ gives

$$(\hat{\theta}_{t-1} - \psi_t)^2 - (\theta - \psi_t)^2 = (\hat{\theta}_{t-1} - \theta)^2 + 2(\theta - \psi_t)(\hat{\theta}_{t-1} - \theta).$$

Summing over $t = 2, \dots, n$, dividing by n , and subtracting the missing $t = 1$ term of B_n :

$$\begin{aligned} \hat{B}_n - B_n &= \frac{1}{n} \sum_{t=2}^n \left[(\hat{\theta}_{t-1} - \psi_t)^2 - (\theta - \psi_t)^2 \right] - \frac{1}{n} (\theta - \psi_1)^2 \\ &= \underbrace{\frac{1}{n} \sum_{t=2}^n (\hat{\theta}_{t-1} - \theta)^2}_{T_1} + \underbrace{\frac{2}{n} \sum_{t=2}^n (\theta - \psi_t)(\hat{\theta}_{t-1} - \theta)}_{T_2} - \underbrace{\frac{1}{n} (\theta - \psi_1)^2}_{T_3}. \quad \square \end{aligned}$$

Remark 2 (Structure of the bias terms). *The term $T_1 \geq 0$ always. It captures the “noise floor”: the accumulated variance of the running estimate $\hat{\theta}_{t-1}$ around the true mean θ . Heuristically,*

$T_1 = \frac{1}{n} \sum_{t=2}^n (\hat{\theta}_{t-1} - \theta)^2$ has expectation $\frac{1}{n} \sum_{t=2}^n \text{Var}(\hat{\theta}_{t-1})$, which decays as $O(\log(n)/n)$ (see the proof of Theorem ??).

The term $T_3 = \frac{1}{n} (\theta - \psi_1)^2$ is the squared initial model calibration error, divided by n ; it has magnitude $O(1/n)$ and enters with a negative sign, partially offsetting T_1 .

The cross term T_2 involves $(\theta - \psi_t)(\hat{\theta}_{t-1} - \theta)$. Neither factor is mean-zero conditional on $\mathcal{F}_{t-1}$ (only $\mathbb{E}[\hat{\theta}_{t-1} - \theta] = 0$ marginally), so T_2 does not vanish in expectation. However, the proof of Theorem ?? already establishes that $\mathbb{E}[|T_2|] = O(1/\sqrt{n})$, which is a worst-case bound; see Appendix A.1 for a refined analysis showing that $|\mathbb{E}[T_2]| = O(\bar{\sigma}^2 \log(n)/N)$ under model calibration.

Empirically, T_1 dominates $|T_2|$ and T_3 by several orders of magnitude across all configurations tested (Appendix A.3, Figure 1).

Step 2: Estimating the noise floor

The dominant bias term T_1 can be rewritten using the martingale structure. Since $\hat{\theta}_{t-1} - \theta = \frac{1}{t-1} \sum_{s=1}^{t-1} \Delta_s$ and the Δ_s have zero mean and are orthogonal (martingale differences), we have

$$\text{Var}(\hat{\theta}_{t-1}) = \frac{1}{(t-1)^2} \sum_{s=1}^{t-1} \mathbb{E}[\Delta_s^2] = \frac{1}{(t-1)^2} \sum_{s=1}^{t-1} \mathbb{E}[v_s], \quad (8)$$

where $v_s := \text{Var}(\Delta_s | \mathcal{F}_{s-1}) = A_s - B_s$ is the per-step conditional variance, with

$$A_s := \frac{1}{N^2} \sum_{i \notin O_{s-1}} \frac{(y_i - \hat{f}^{(s-1)}(x_i))^2}{q_s(i)}, \quad B_s := \frac{1}{N^2} \left(\sum_{i \notin O_{s-1}} (y_i - \hat{f}^{(s-1)}(x_i)) \right)^2. \quad (9)$$

We therefore need an estimator of $\mathbb{E}[v_s]$ at each step s . A natural candidate is ζ_s^2 , the squared AIPW correction from (2). The following lemma justifies this choice.

Lemma 3. For each $s = 1, \dots, n$,

$$\mathbb{E}[\zeta_s^2 | \mathcal{F}_{s-1}] = A_s. \quad (10)$$

Consequently, ζ_s^2 is an unbiased estimator of $A_s = v_s + B_s$, overestimating v_s by $B_s \geq 0$.

Proof. By definition of ζ_s and $q_s(\cdot) = P(I_s = \cdot | \mathcal{F}_{s-1})$,

$$\mathbb{E}[\zeta_s^2 | \mathcal{F}_{s-1}] = \sum_{i \notin O_{s-1}} q_s(i) \cdot \frac{1}{N^2} \frac{(y_i - \hat{f}^{(s-1)}(x_i))^2}{q_s(i)^2} = \frac{1}{N^2} \sum_{i \notin O_{s-1}} \frac{(y_i - \hat{f}^{(s-1)}(x_i))^2}{q_s(i)} = A_s \square$$

Remark 4 ($B_s \ll A_s$ in practice). The overestimation B_s/A_s is a property of model calibration, not of the sample size. Specifically, $B_s = \frac{1}{N^2} (M_s \bar{r}_s)^2$ and $A_s \geq \frac{1}{N^2} \sum_{i \notin O_{s-1}} (y_i - \hat{f}^{(s-1)}(x_i))^2 = \frac{M_s}{N^2} \bar{r}_s^2$ (with equality for uniform sampling), where $M_s = N - s + 1$, $\bar{r}_s$ is the mean residual, and $\bar{r}_s^2$ is the mean squared residual over unobserved items. Thus $B_s/A_s \leq \bar{r}_s^2 / \bar{r}_s^2$, which is small whenever the ML model is approximately calibrated ($|\bar{r}_s| \ll \sqrt{\bar{r}_s^2}$). See Appendix A.2 for a detailed analysis and empirical verification (Figure 2).

Step 3: Construction of $\hat{C}_n$

We now define the correction term by “plugging in” the estimator ζ_s^2 for v_s in the expression for $\mathbb{E}[T_1]$.

Proposition 5 (Bias correction). *Define*

$$\hat{C}_n := \frac{1}{n} \sum_{t=2}^n \frac{1}{(t-1)^2} \sum_{s=1}^{t-1} \zeta_s^2. \quad (11)$$

Then:

(a) (**Closed form**). *Exchanging the order of summation,*

$$\hat{C}_n = \frac{1}{n} \sum_{s=1}^{n-1} \zeta_s^2 w_s, \quad w_s := \sum_{k=s}^{n-1} \frac{1}{k^2}. \quad (12)$$

(b) (**Expectation**). $\mathbb{E}[\hat{C}_n] = \mathbb{E}[T_1] + R_C$, where $R_C := \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[B_s] w_s \geq 0$.

Proof. (a). The double sum in (11) runs over pairs (s, t) with $1 \leq s \leq t-1$ and $2 \leq t \leq n$, equivalently $1 \leq s \leq n-1$ and $s+1 \leq t \leq n$. Exchanging the order and substituting $k = t-1$:

$$\sum_{t=2}^n \frac{1}{(t-1)^2} \sum_{s=1}^{t-1} \zeta_s^2 = \sum_{s=1}^{n-1} \zeta_s^2 \sum_{t=s+1}^n \frac{1}{(t-1)^2} = \sum_{s=1}^{n-1} \zeta_s^2 \sum_{k=s}^{n-1} \frac{1}{k^2}.$$

(b). By the tower property and Lemma 3, $\mathbb{E}[\zeta_s^2] = \mathbb{E}[A_s] = \mathbb{E}[v_s] + \mathbb{E}[B_s]$. Thus,

$$\mathbb{E}[\hat{C}_n] = \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[A_s] w_s = \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[v_s] w_s + \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[B_s] w_s.$$

It remains to identify the first sum with $\mathbb{E}[T_1]$. By (8),

$$\mathbb{E}[T_1] = \frac{1}{n} \sum_{t=2}^n \text{Var}(\hat{\theta}_{t-1}) = \frac{1}{n} \sum_{t=2}^n \frac{1}{(t-1)^2} \sum_{s=1}^{t-1} \mathbb{E}[v_s].$$

Exchanging the order of summation (identical to part (a) with $\mathbb{E}[v_s]$ in place of ζ_s^2):

$$\mathbb{E}[T_1] = \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[v_s] w_s.$$

Hence $\mathbb{E}[\hat{C}_n] = \mathbb{E}[T_1] + R_C$ with $R_C = \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[B_s] w_s \geq 0$ since $B_s \geq 0$ and $w_s > 0$. $\square$

Step 4: The corrected estimator and its properties

Theorem 6 (Bias-corrected variance estimator). *Define*

$$\hat{\sigma}_{\text{cor}}^2 := \hat{A}_n - \hat{B}_n + \hat{C}_n. \quad (13)$$

Under the assumptions of Theorem ??:

(a) (**Bias cancellation**). *The expectation of the corrected estimator satisfies*

$$\mathbb{E}[\hat{\sigma}_{\text{cor}}^2] - \mathbb{E}[A_n - B_n] = R_C - \mathbb{E}[T_2] + \mathbb{E}[T_3], \quad (14)$$

where the dominant original bias $\mathbb{E}[T_1]$ has been exactly cancelled. The remaining terms satisfy $R_C \geq 0$, $\mathbb{E}[T_3] \geq 0$, and $|\mathbb{E}[T_2]| = O(1/\sqrt{n})$ (worst case) or $O(\bar{\sigma}^2 \log(n)/N)$ under model calibration. In particular, the corrected estimator is slightly conservative: its CIs are wider than necessary, ensuring coverage at or above the nominal level.

(b) (**Consistency**). $\hat{\sigma}_{\text{cor}}^2 \xrightarrow{P} \sigma^2$.

(c) (**Asymptotic normality**). The conclusion of Theorem ?? holds with $\hat{\sigma}_{\text{cor}}$ in place of $\hat{\sigma}_n$:

$$\sqrt{n}(\hat{\theta}_n - \theta) / \hat{\sigma}_{\text{cor}} \xrightarrow{D} \mathcal{N}(0, 1). \quad (15)$$

Proof. (a). By the pathwise decomposition $\hat{B}_n = B_n + T_1 + T_2 - T_3$ (Proposition 1),

$$\begin{aligned} \mathbb{E}[\hat{\sigma}_{\text{cor}}^2] &= \mathbb{E}[\hat{A}_n] - \mathbb{E}[\hat{B}_n] + \mathbb{E}[\hat{C}_n] \\ &= \mathbb{E}[A_n] - (\mathbb{E}[B_n] + \mathbb{E}[T_1] + \mathbb{E}[T_2] - \mathbb{E}[T_3]) + (\mathbb{E}[T_1] + R_C), \end{aligned}$$

where we used that $\mathbb{E}[\hat{A}_n] = \mathbb{E}[A_n]$ (the martingale difference property of $C_t = \zeta_t^2 - A_t$ established in the proof of Theorem ?? gives $\mathbb{E}[\hat{A}_n - A_n] = 0$), and Proposition 5(b) for $\mathbb{E}[\hat{C}_n]$. The $\mathbb{E}[T_1]$ terms cancel, yielding (14).

(b). Since $\hat{\sigma}_{\text{cor}}^2 = \hat{\sigma}_n^2 + \hat{C}_n$ and $\hat{\sigma}_n^2 \xrightarrow{P} \sigma^2$ (Theorem ??), it suffices to show $\hat{C}_n \xrightarrow{P} 0$. We have

$$0 \leq \hat{C}_n = \frac{1}{n} \sum_{s=1}^{n-1} \zeta_s^2 w_s \leq \frac{w_1}{n} \sum_{s=1}^{n-1} \zeta_s^2 = w_1 \cdot \frac{n-1}{n} \cdot \hat{A}'_{n-1} \leq w_1 \cdot \hat{A}_n,$$

where $\hat{A}'_n := \frac{1}{n-1} \sum_{s=1}^{n-1} \zeta_s^2$. Now $w_1 = \sum_{k=1}^{n-1} \frac{1}{k^2} \leq \frac{\pi^2}{6}$, but more importantly $w_1 = O(1)$ while $\hat{A}_n \xrightarrow{P} A := \lim A_n$, so this bound is $O(1)$, which is not sufficient by itself.

Instead, we bound directly: $\hat{C}_n = \frac{1}{n} \sum_{s=1}^{n-1} \zeta_s^2 w_s$. Since $w_s = \sum_{k=s}^{n-1} 1/k^2 \leq 1/s + \int_s^\infty x^{-2} dx = 2/s$, we get

$$\hat{C}_n \leq \frac{2}{n} \sum_{s=1}^{n-1} \frac{\zeta_s^2}{s}.$$

Taking expectations: $\mathbb{E}[\hat{C}_n] \leq \frac{2}{n} \sum_{s=1}^{n-1} \frac{\mathbb{E}[A_s]}{s}$. Under the moment conditions of Theorem ??, $\mathbb{E}[A_s] \leq \mathbb{E}[\zeta_s^2] \leq (\mathbb{E}[\zeta_s^4])^{1/2} \leq C^{1/2}$ by Assumption ?? and Jensen's inequality. Therefore,

$$\mathbb{E}[\hat{C}_n] \leq \frac{2C^{1/2}}{n} \sum_{s=1}^{n-1} \frac{1}{s} = O\left(\frac{\log n}{n}\right) \rightarrow 0.$$

Since $\hat{C}_n \geq 0$, this implies $\hat{C}_n \xrightarrow{P} 0$ by Markov's inequality.

(c). Since $\hat{\sigma}_{\text{cor}}^2 \xrightarrow{P} \sigma^2 > 0$, we have $\hat{\sigma}_{\text{cor}} \xrightarrow{P} \sigma$, and Slutsky's theorem applied to the result $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} \mathcal{N}(0, \sigma^2)$ from Theorem ?? gives (15). $\square$

Remark 7 (Implementation). The correction (11) requires minimal overhead. At each step $t \geq 2$, the running sum $S_t := \sum_{s=1}^{t-1} \zeta_s^2$ is already maintained for computing $\hat{A}_n$. Before appending ζ_t^2 to this sum, one accumulates $\hat{C}_n \leftarrow \hat{C}_n + S_t/(t-1)^2$. After all n rounds, divide by n . The cost is one addition per step.

Remark 8 (Comparison with the “A-only” estimator). A simpler alternative is to drop $\hat{B}_n$ entirely, using $\hat{\sigma}_A^2 := \hat{A}_n$. This estimator is always conservative since $\hat{A}_n$ estimates $A_n \geq A_n - B_n$. However, it overestimates the variance by B_n on average, which does not vanish—unlike the residual R_C of the corrected estimator, which is $O(\log(n)/n)$ and multiplied by the small ratio B_s/v_s . The corrected estimator $\hat{\sigma}_{\text{cor}}^2$ is strictly more accurate than $\hat{\sigma}_A^2$ while retaining the conservative property.

A Supplementary Material for the Bias Correction

A.1 Refined bounds on T_2 and T_3

We provide tighter bounds on $\mathbb{E}[T_2]$ and $\mathbb{E}[T_3]$ that complement the worst-case rates established in the proof of Theorem ??.

Bound on $\mathbb{E}[T_3]$

Since $\psi_1 = \frac{1}{N} \sum_{i=1}^N \hat{f}^{(0)}(x_i)$ depends only on the initial (pre-data) model, T_3 is deterministic:

$$T_3 = \frac{1}{n} (\theta - \psi_1)^2 = \frac{1}{n} \left(\frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}^{(0)}(x_i)) \right)^2.$$

This is the squared mean residual of the initial model, divided by n . By the Cauchy–Schwarz–Jensen chain used in the proof of Theorem ??, $T_3 \leq C^{1/2}/n = O(1/n)$.

More precisely, $T_3 = \bar{r}_1^2/n$ where $\bar{r}_1$ is the initial model’s mean residual. For a well-calibrated model (e.g., a Bayesian logistic regression fitted by maximum likelihood), $|\bar{r}_1|$ is typically of order 10^{-2} or smaller, so $T_3 \sim 10^{-4}/n$. At $n = 3000$, this is $O(10^{-8})$.

Bound on $\mathbb{E}[T_2]$

The proof of Theorem ?? establishes $\mathbb{E}[|T_2|] = O(1/\sqrt{n})$ via Cauchy–Schwarz. This worst-case bound does not exploit the structure of $\theta - \psi_t$. We now show that this quantity is small whenever the ML model has small mean residual—a model calibration property rather than a sample-size condition.

Recall that $\theta - \psi_t = \frac{1}{N} \sum_{i \notin O_{t-1}} (y_i - \hat{f}^{(t-1)}(x_i))$ (the mean residual over unobserved items, times $(N - t + 1)/N$). Define $M_t := N - t + 1 = |R_{t-1}|$ and $\bar{r}_t := \frac{1}{M_t} \sum_{i \notin O_{t-1}} (y_i - \hat{f}^{(t-1)}(x_i))$, so that $\theta - \psi_t = (M_t/N) \bar{r}_t$.

The contribution to T_2 at step t can also be written as:

$$(\theta - \psi_t)(\hat{\theta}_{t-1} - \theta) = \frac{M_t}{N} \bar{r}_t \cdot (\hat{\theta}_{t-1} - \theta).$$

To bound $\mathbb{E}[T_2]$, note that each summand is a product of two terms, both of which involve shared randomness through O_{t-1} . We decompose the covariance structure as follows. Since $\hat{\theta}_{t-1} = \theta + \frac{1}{t-1} \sum_{s=1}^{t-1} \Delta_s$ and $\psi_t = \psi_1 + \sum_{s=1}^{t-1} (\psi_{s+1} - \psi_s)$ where the increments $\psi_{s+1} - \psi_s$ depend on I_s (the item sampled at step s , which replaces $\hat{f}^{(s-1)}(x_{I_s})$ with y_{I_s} and may update the model), both terms share the common randomness from sampling $I_1, \dots, I_{t-1}$.

The key observation is that the “effective covariance per step” between ζ_s/N (contributing to $\hat{\theta}$ via $1/q_s$ weighting) and r_{I_s}/N (contributing to $\psi_{s+1} - \psi_s$ at unit weight) is of order $\bar{\sigma}^2/N$: the common randomness is amplified by $1/q_s$ in the estimator but only by 1 in the plug-in, creating an asymmetry that makes the correlation weak when N is large relative to n . This propagates through the running average $\hat{\theta}_{t-1}$ with harmonic weights $1/(t-1)$, giving:

$$|\mathbb{E}[T_2]| \lesssim \frac{\bar{\sigma}^2 \log n}{N}, \tag{16}$$

where $\bar{\sigma}^2 = \frac{1}{n} \sum_s \mathbb{E}[v_s]$ is the average per-step conditional variance.

Remark 9. The bound (16) should be understood as a heuristic motivated by the conditional covariance structure; a fully rigorous proof would require tracking the joint distribution of $(\hat{\theta}_{t-1}, \psi_t)$ through the adaptive sampling process, which is complicated by the model updates. We verify empirically that $|\mathbb{E}[T_2]|/|\mathbb{E}[T_1]|$ is of order 10^{-3} or smaller across all configurations (Appendix A.3).

A.2 The ratio B_s/A_s

The residual $R_C = \frac{1}{n} \sum_{s=1}^{n-1} \mathbb{E}[B_s] w_s$ from Proposition 5 is controlled by the per-step ratio B_s/A_s . We analyze this ratio.

For step s , let $M_s = N - s + 1$, and let $r_i := y_i - \hat{f}^{(s-1)}(x_i)$ denote the residuals for $i \notin O_{s-1}$. Define

$$\bar{r}_s := \frac{1}{M_s} \sum_{i \notin O_{s-1}} r_i, \quad \overline{r_s^2} := \frac{1}{M_s} \sum_{i \notin O_{s-1}} r_i^2.$$

Then:

$$B_s = \frac{M_s^2 \bar{r}_s^2}{N^2}, \quad A_s = \frac{1}{N^2} \sum_{i \notin O_{s-1}} \frac{r_i^2}{q_s(i)} \geq \frac{1}{N^2} \sum_{i \notin O_{s-1}} r_i^2 = \frac{M_s \overline{r_s^2}}{N^2},$$

where the inequality uses $q_s(i) \leq 1$ for all i . Hence:

$$\frac{B_s}{A_s} \leq \frac{M_s^2 \bar{r}_s^2}{M_s \overline{r_s^2}} = \frac{M_s \bar{r}_s^2}{\overline{r_s^2}} = \frac{M_s \bar{r}_s^2}{\sigma_{r,s}^2 + \bar{r}_s^2}, \quad (17)$$

where $\sigma_{r,s}^2 := \overline{r_s^2} - \bar{r}_s^2 \geq 0$ is the variance of residuals across the remaining unobserved items. Note that the N^2 factors cancel completely, so the ratio depends only on the moments of the residual distribution, not on N directly.

For active sampling with $\tau < 1$, items with large $|r_i|$ receive $q_s(i)$ above the uniform level $1/M_s$, which increases A_s relative to the uniform case without affecting B_s . This further decreases B_s/A_s .

When is B_s/A_s small?

The ratio (17) is small when the mean residual $\bar{r}_s$ is small relative to the root-mean-square residual $\sqrt{\overline{r_s^2}}$. This is a model calibration property:

- For binary outcomes $y_i \in \{0, 1\}$ with probabilistic predictions $\hat{f}^{(s-1)}(x_i) \in [0, 1]$, the mean squared residual $\overline{r_s^2}$ equals the Brier score, typically ~ 0.1 – 0.25 . The mean residual $\bar{r}_s = \bar{y}_s - \bar{\hat{f}}_s$ is the calibration error, which is $|\bar{r}_s| \lesssim 0.01$ for a model fitted by maximum likelihood.
- The ratio thus satisfies $B_s/A_s \leq M_s \bar{r}_s^2 / \overline{r_s^2} \lesssim M_s \cdot (0.01)^2 / 0.1 = 10^{-3} M_s$.

This bound scales with M_s and will not be small in general without further assumptions. However, the relevant quantity for R_C is $\mathbb{E}[B_s]/\mathbb{E}[v_s]$, since

$$\frac{R_C}{\mathbb{E}[T_1]} = \frac{\sum_{s=1}^{n-1} \mathbb{E}[B_s] w_s}{\sum_{s=1}^{n-1} \mathbb{E}[v_s] w_s} \leq \max_{1 \leq s \leq n-1} \frac{\mathbb{E}[B_s]}{\mathbb{E}[v_s]}.$$

We verify empirically that $B_s/v_s \lesssim 0.5\%$ at all steps and budgets in both datasets tested (Figure 2). Combined with $\mathbb{E}[T_1]/\bar{\sigma}^2 \leq 0.22$ (at the largest budget), this gives $R_C/\bar{\sigma}^2 \leq 0.005 \times 0.22 \approx 10^{-3}$, which translates to a coverage overcorrection of $\lesssim 0.0001$ —undetectable at 100 seeds.

A.3 Empirical verification

We verify the two key empirical claims: (i) T_1 dominates $|T_2|$ and T_3 , and (ii) $B_s \ll A_s$ at every step. The verification procedure (implemented in `verify_bias_decomposition.py`) runs the full WOR-PAI loop on each (dataset, budget) combination with best-tuned hyperparameters, tracking the per-step quantities A_s , B_s and the cumulative T_1 , T_2 , T_3 over 100 seeds $\times$ all test models.

[Placeholder: bias decomposition results from `verify_bias_decomposition.py`]

Figure 1: Empirical verification of $T_1 \gg |T_2|, T_3$. Top: Normalized quantities $T_1/\bar{\sigma}^2$, $|T_2|/\bar{\sigma}^2$, $T_3/\bar{\sigma}^2$ as a function of budget. Bottom: Ratio $|T_2|/T_1$ and T_3/T_1 .

[Placeholder: B_s/A_s profiles from `verify_bias_decomposition.py`]

Figure 2: Per-step ratio B_s/A_s as a function of sampling step s , averaged over 100 seeds and all test models. Left: MMLU-Pro at 25% budget. Right: BBH+GPQA+IFEval+MATH+MuSR at 25% budget. The ratio remains below 0.5% throughout.